RAJAT MURHE

Final Year Computer Science Undergraduate
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PROFILE

Final year Computer Science undergraduate (BTech, AI and Machine Learning, VIT Chennai) with three internships, who builds **multi agent systems** and the pieces around them: LangGraph workflows with MCP tools, state and conversational context, human review steps, and eval harnesses with red team sets that run in CI. I have connected agents to real systems such as Google Calendar and third party APIs, and I learned most from debugging agents that broke in live testing.

EXPERIENCE

AI Product Growth and Strategy Intern, DCS AI Technologies Sep 2026 to Present

- Integrated **third party APIs** into AI powered product workflows, connecting external services with application features and validating end to end functionality.
- Worked on a platform's multimodal AI capabilities, testing and validating image, video, audio and product ad generation workflows, including API responses and **credit consumption**.
- Contributed to an AI connector platform's integration workflows by testing connectors, dashboard operations, execution flows and **role based permissions**.
- **Debugged** integration issues and documented test cases and findings to improve product reliability and release readiness.

AI/ML Intern, Linkific | Remote Jun 2026 to Aug 2026

- Built the **backend** for a Secure Enterprise AI Assistant that answers questions from company documents, using FastAPI, RAG on PostgreSQL with pgvector and a Groq LLM.
- Added **role based access control** to the API so people only get answers from documents they are allowed to see, and helped with troubleshooting and debugging for users.

AI Intern, Rungta Ventures AI Jan 2026 to May 2026

- Built a **semantic search** feature on MyCA using embeddings, and wrote Python scripts with NLTK and spaCy that validate uploaded documents, with a team of around ten people on a live product that uses Firebase for OTP login.

PROJECTS

ClinicVoice AI, Voice Receptionist for Clinics | GitHub Jun 2025 to Sep 2025

Python* | *FastAPI* | *Node.js* | *LangGraph* | *MCP* | *Groq* | *Amazon Bedrock* | *SQLite* | *Docker Compose* | *GitHub Actions

- Built a voice AI agent system with **LangGraph and MCP tools**, with six agents where only the FAQ agent calls an LLM, that answers patient questions, books, cancels and reschedules real appointments through Google Calendar and refuses medical advice.
- Added four safety layers and a **global safety gate** after live testing showed a gap, plus Amazon Bedrock as a second LLM provider behind one interface.
- Built an eval harness with a red team set, fixed the failures it found, and ran it in **GitHub Actions CI** on every push as an automated regression check.

Atlas, AI Powered Commercial Intelligence Workbench | GitHub

Python* | *SQL* | *PostgreSQL* | *LangGraph* | *FastAPI* | *Next.js* | *TypeScript* | *Redis* | *Docker* | *GitHub Actions

- Developed a **LangGraph analytics agent** with conversational context, SQL validation, restricted database access, query timeouts and evidence backed responses, over a PostgreSQL star schema of 24,576 synthetic pharmaceutical sales records.
- Built forecasting and analytics that reached 1.25% forecast MAPE versus 4.70% for a seasonal naive baseline, with eight data quality checks in a reproducible ETL.
- Delivered a Next.js dashboard with interactive charts and SQL inspection, and verified the Docker stack and **19 backend tests** with GitHub Actions checks.

FieldProof AI, Field Service Video QA Platform | GitHub Aug 2026 to Sep 2026

Python* | *FastAPI* | *Next.js* | *TypeScript* | *PostgreSQL* | *Gemini* | *Clerk

- Built a multi tenant platform where **Gemini** checks technician job videos against a checklist and a person reviews and approves each finding, with an audit trail of every job step.
- Designed a PostgreSQL **job queue** with retries and stale job recovery, so long video checks run on their own without the frontend open.

AI Job Assistant, Resume and Job Match Tool | GitHub Jan 2026 to Aug 2026

Next.js* | *React* | *TypeScript* | *Redux Toolkit* | *FastAPI* | *PostgreSQL* | *pgvector* | *n8n* | *Gemini* | *Ollama* | *OpenAI APIs

- Built an AI job search assistant that reads a resume and a job description, shows how well they match with **evidence** for each matched and missing skill, and drafts tailored resumes, cover letters and emails.
- Built n8n workflows that automate the application steps, with a multi provider **LLM API** layer that checks model output against a schema.

TECHNICAL SKILLS

Agents and Orchestration: LangGraph, LangChain, MCP, tool calling, multi agent systems, human in the loop review, structured outputs, RAG, embeddings, pgvector, OpenAI, Claude, Gemini, Groq, Ollama, Amazon Bedrock, n8n

Evals and Reliability: eval harness, red team sets, regression tests in CI, prompt injection tests, safety guardrails, retries and stale job recovery, audit trails

Backend and Frontend: Python, FastAPI, Node.js, PostgreSQL, SQLite, Redis, REST APIs, Docker Compose, GitHub Actions CI, Git, React, Next.js, TypeScript

CS Fundamentals: Data structures and algorithms (DSA), DBMS, object oriented programming (OOPs), Python, Java, C++, Go

EDUCATION

Vellore Institute of Technology, Chennai | BTech, Computer Science, Specialization in AI and Machine Learning Aug 2023 to Sep 2027

CERTIFICATIONS AND ACHIEVEMENTS

- Won **First Prize** at the AGRHINOVA Hackathon, VIT Chennai, for a crop diagnostics platform that combines AI image classification with a multilingual RAG assistant.
- AWS Certified Developer Associate. AWS Cloud Practitioner. Google Cloud Professional ML Engineer. Microsoft Azure Developer Associate, Azure Fundamentals, ML Operations Engineer Associate. IBM Advanced Generative AI. Avery Dennison Scholarship.

LEADERSHIP AND EXTRACURRICULAR

- Vice President, Swarajya Marathi Literary Association (2025 to 2026). State Level Cricketer, Maharashtra Under 16.